



Session 1

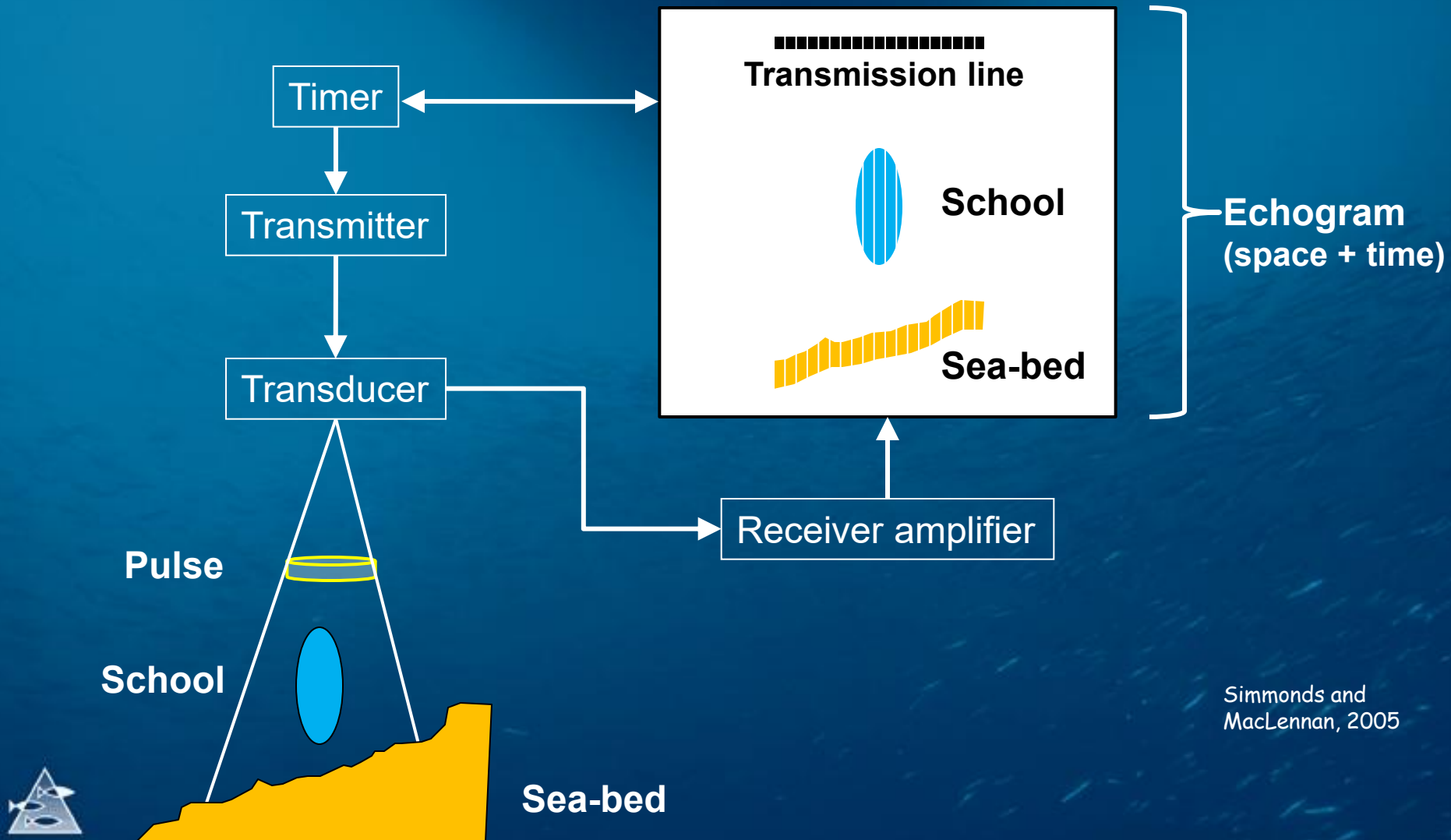
Basic theory and operation of fisheries sonar

Interpretation of acoustic data from fisheries sonar



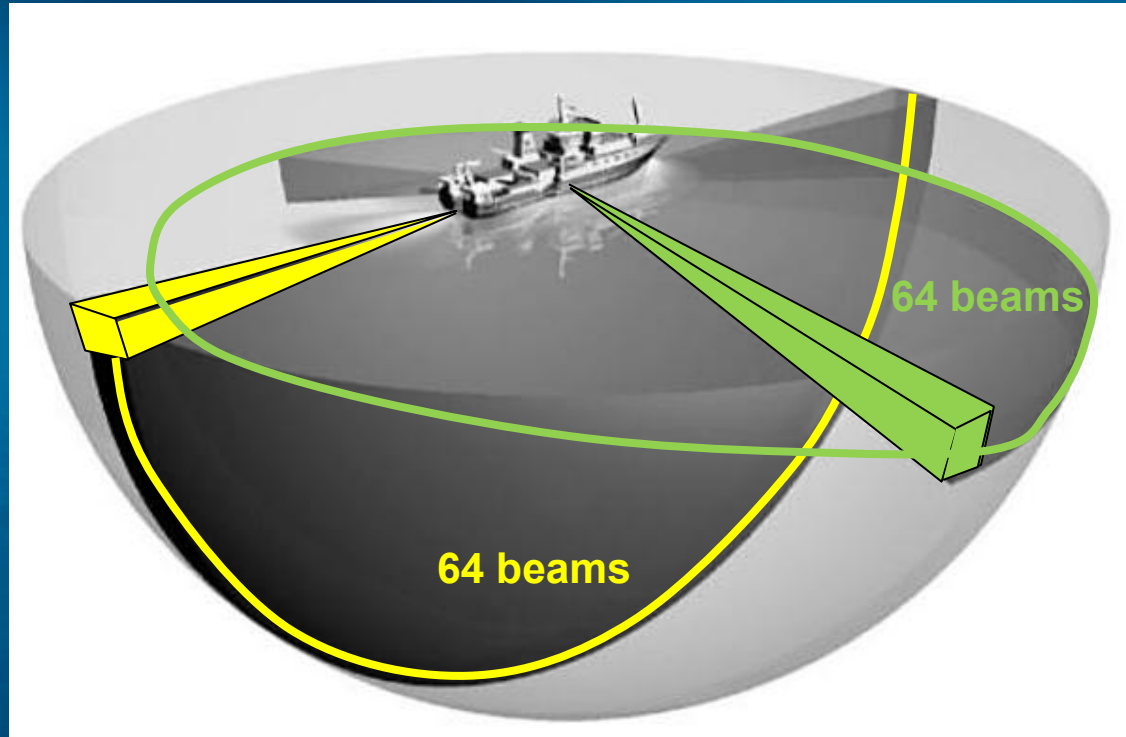
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Concept of echosounding



Omni directional sonar

- Multibeam sonar
- Frequencies: 14 to 114 kHz
- max. 8000 m range
- Horizontal and vertical beam fans

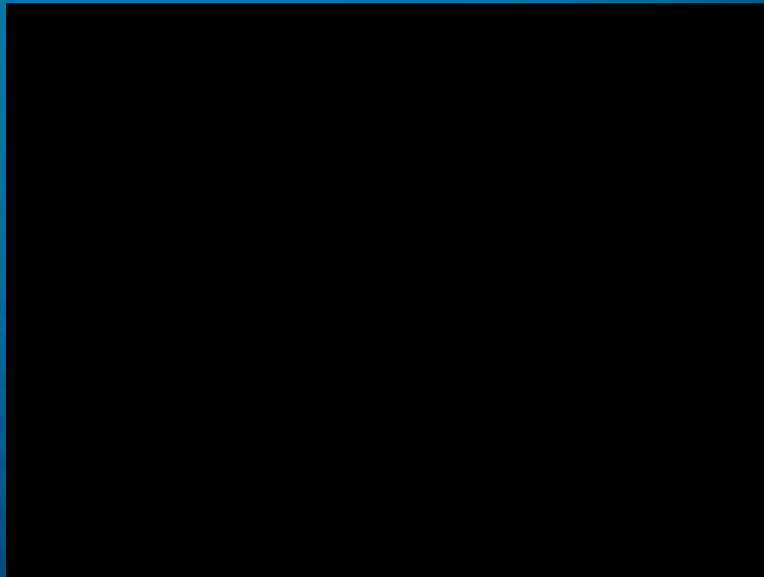


Simmonds and MacLennan, 2005



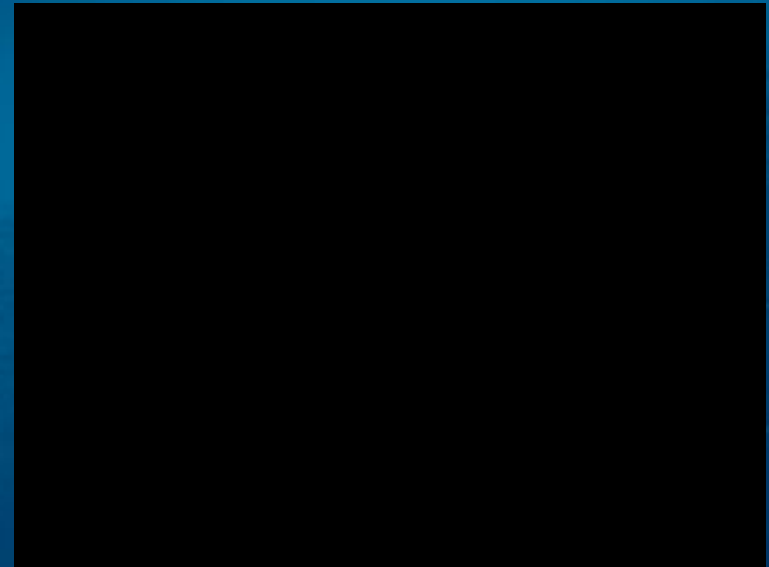
Operation of omni-directional sonar

Omni transmission at varying tilt angles



Simrad

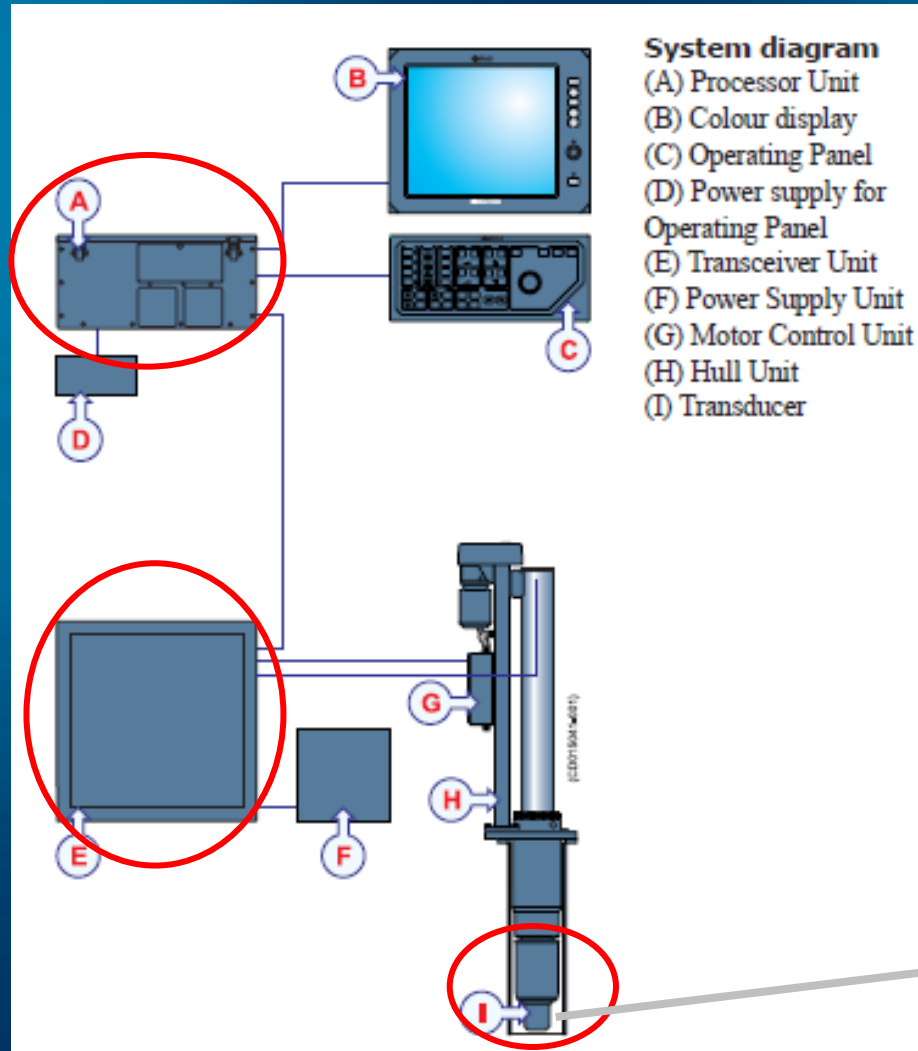
Omni and vertical transmission



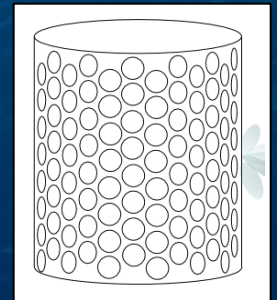
Simrad



Sonar components

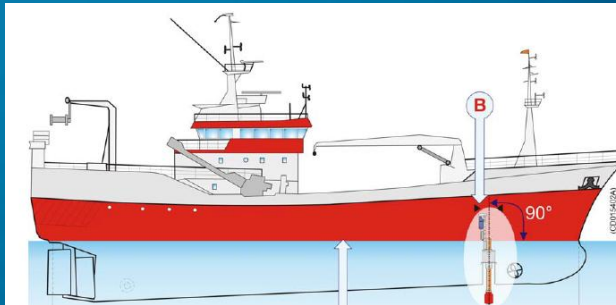
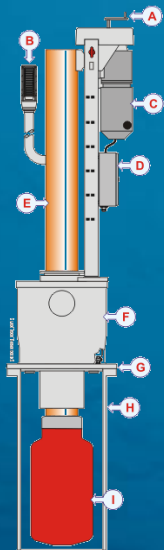


**Cylindrical transducer
256 elements (SX90)**



Sonar components

Recomended installation



Cylindrical transducer lowered up to 1.6 m below vessel hull



High/medium frequency sonar (i. e. SH90, SC90, FSV-85)

Frequency: 114 kHz (85 kHz)

Range: 50 to 2000 m

Tilt: $+10^{\circ}$ to -60°

Pulse modes: CW and FM

Transmission modes:

- 360° omnidirectional
- 180° vertical

Low frequency sonar (i. e. ST90, SU90, FSV-25)

Frequency: 14 to 30 kHz

Range: 50 to 8000 m

Tilt: $+10^{\circ}$ to -60°

Pulse modes: CW and FM

Transmission modes:

- 360° omnidirectional
- 180° vertical



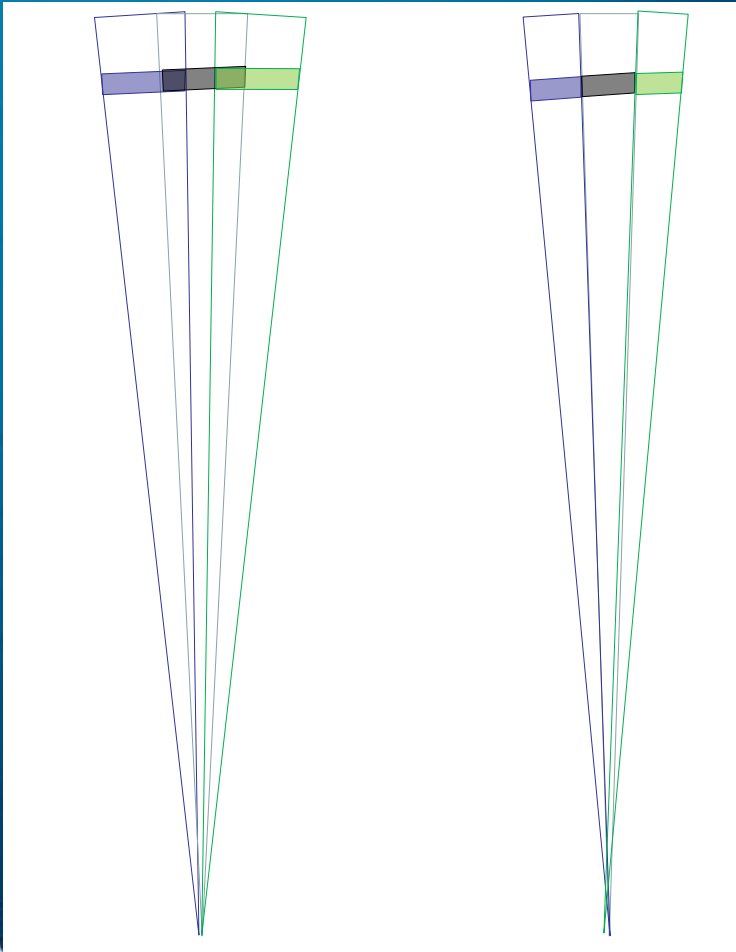
Short range
High resolution

Long range
Low resolution

Horizontal opening angles

Real overlapping beams
10 deg angle

Graphical view
6 deg angle (360°/64 beams)



Estimated horizontal opening angles by frequency (SU90)

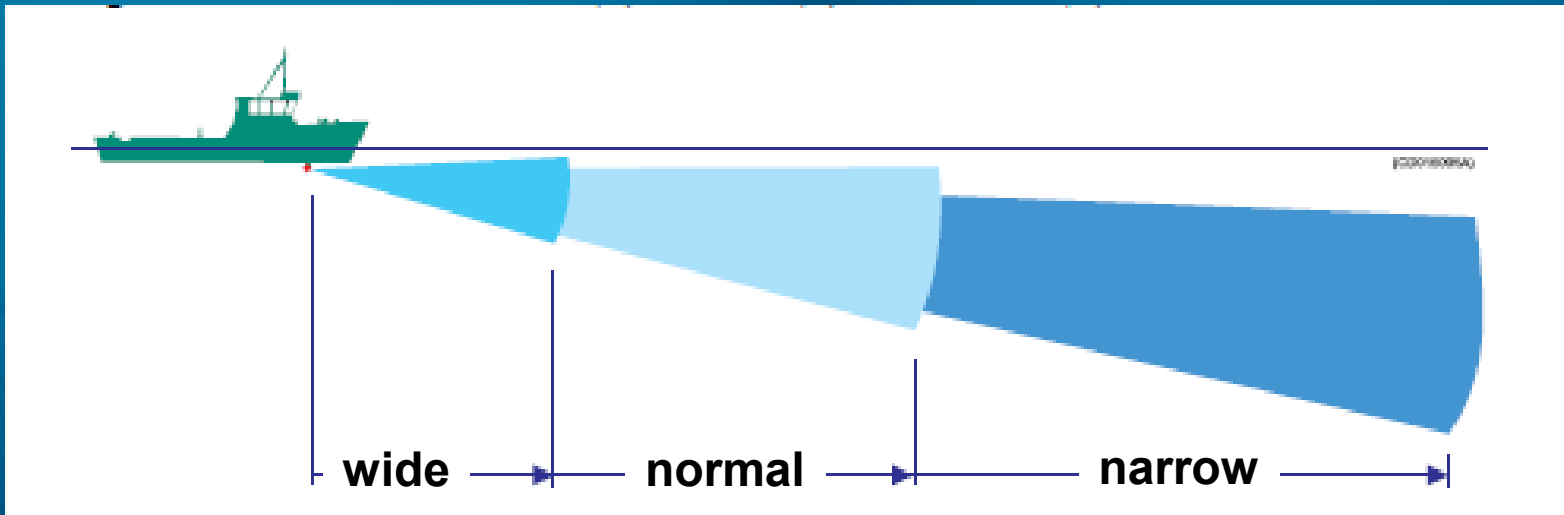
<u>angle</u>	<u>freq</u>
13	20
12	21
12	22
11	23
11	24
10	25
10	26
9	27
9	28
8	29
8	30

Horizontal beam angle = $13 \cdot 20 / \text{freq}$

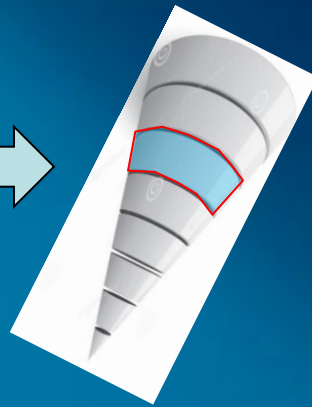
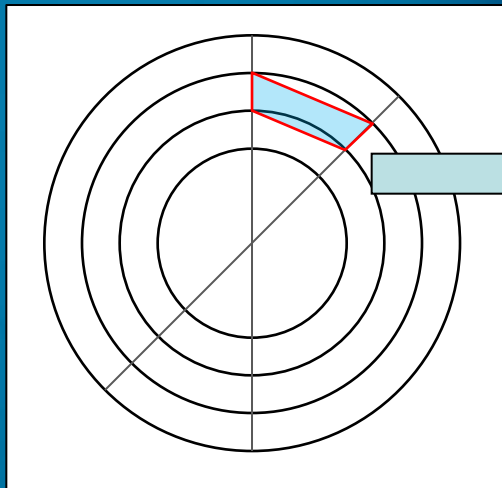


Vertical opening angles

- At 17 kHz normal beam width 6.5° (ST90)
- At 30 kHz normal beam width 7.3° (SX90)
- At 30 kHz normal beam width 5.2° (SU90)
- At 114 kHz normal beam width 7.5° (SH90)



Real size of sample cell in 100 m range in 500 m total range beam



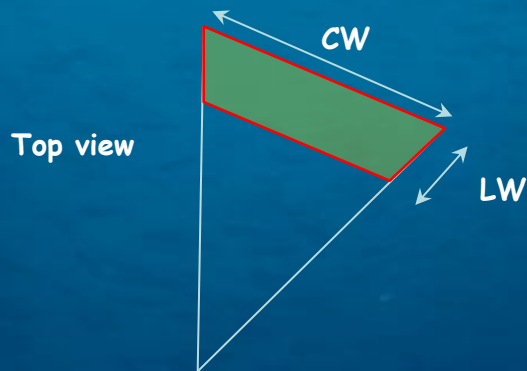
Top view

Side view

LW = 0.2 m

Front view

CW = 10 m



Top view

$CW = 10 \text{ m}$ $LW = 0.2 \text{ m}$

$Z = 27 \text{ m}$



Side view

$Z = 27 \text{ m}$

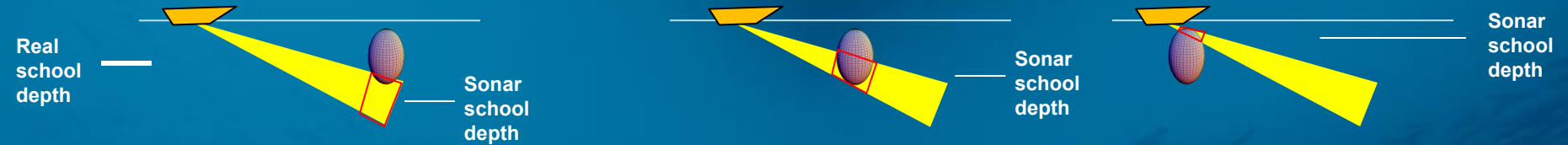


Sonar 360 deg sampling mode

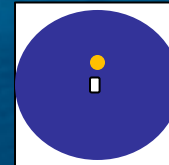
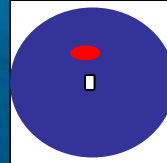
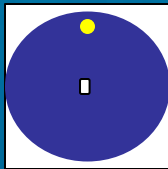
time= 0

time= 1

time= 2



Sonar
horizontal
view



Low backscattering

High backscattering

Medium backscattering

Small school coverage
Large sample bin

Large school coverage
Medium sample bin

Small school coverage
Small sample bin

Real school depth < Sonar school depth

Real school depth $\approx$ Sonar school depth

Real school depth > Sonar school depth



Pulse duration (milliseconds, ms) at different ranges and pulse form (i.e SU90)

CW and FM pulse lengths for different ranges							
Range (m)	CW Short	CW Normal	CW Long	FM Auto	FM Short	FM Normal	FM Long
150	1	2	6	6	2	4	6
300	2	4	12	12	4	8	12
450	3	6	17	18	6	12	18
600	4	8	24	24	8	16	24
900	6	12	34	36	12	24	36
1200	7	14	41	48	16	32	48
1500	8	17	51	60	20	40	60
2000	12	23	68	80	27	54	80
2500	14	29	85	85	33	67	85
3000	17	34	85	85	40	80	85
3500	20	40	85	85	47	85	85
4500	25	51	85	85	60	85	85

Pulse duration EK60 = 1.021 ms

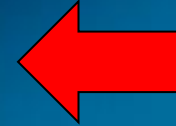
Shorter pulse length improves resolution



Many combinations of pulse form, pulse length at different ranges

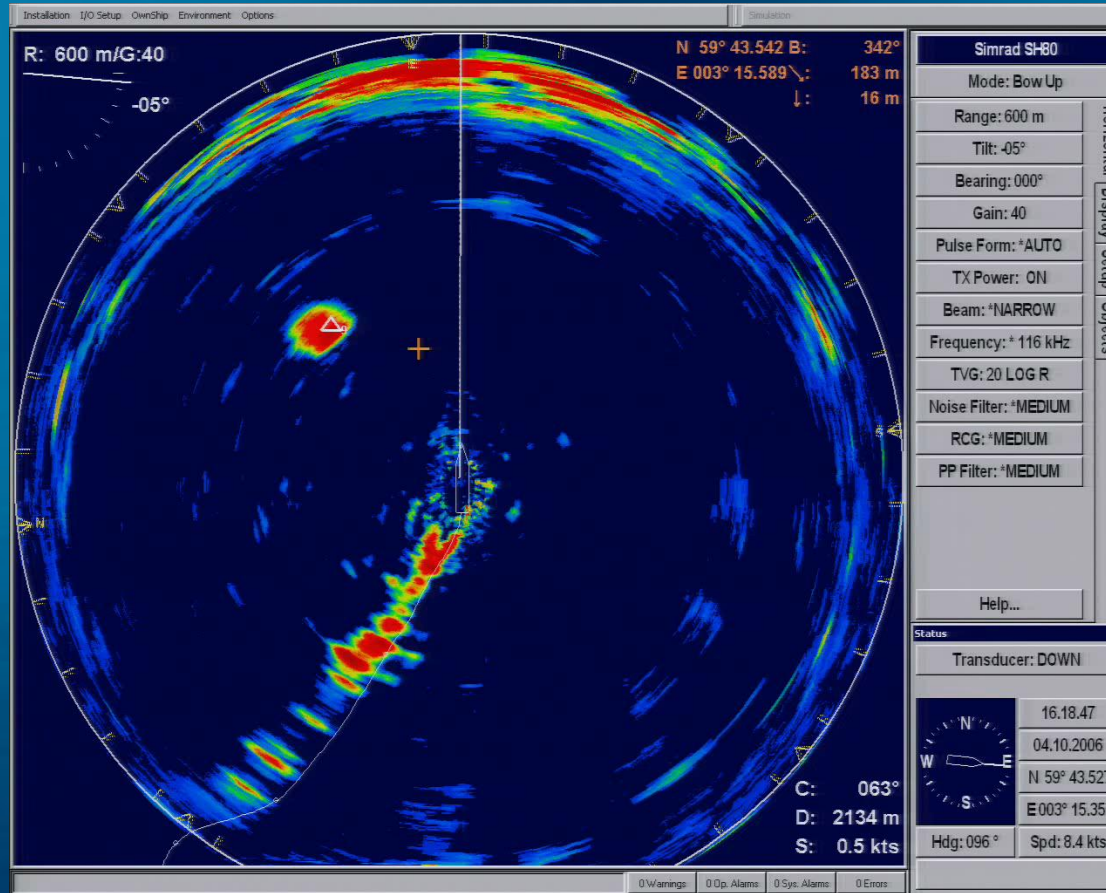
Main problems using omni sonars

- Surface reverberation
 - wind, breaking waves, turbulence
- Bottom echoes
 - when sampling demersal species
 - pelagic species inside fjords and shallow waters
- Vessel generated noise
 - machinery, bubbles below hull
- Biological
 - Unwanted targets (other species, plankton)



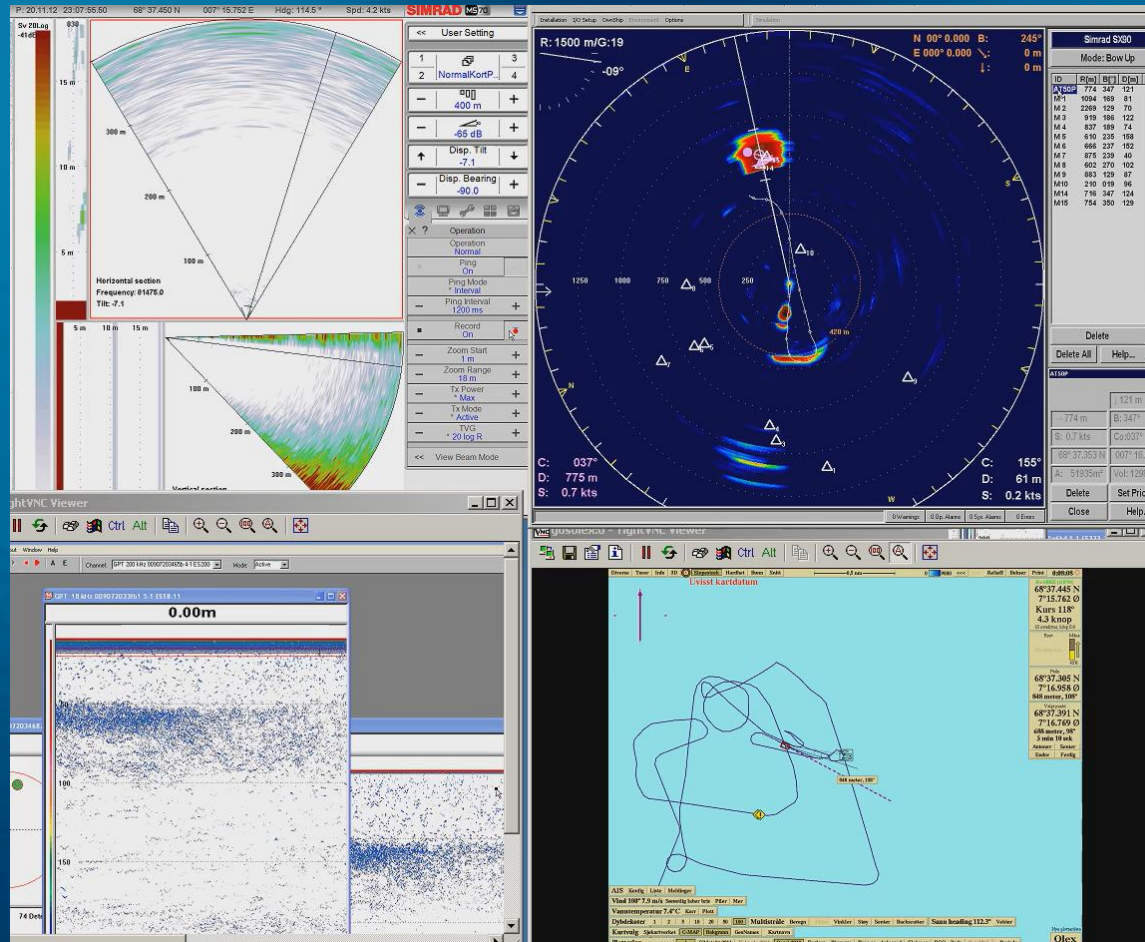
Examples of sonar data

SH80 data of mackerel school during fishing 2006



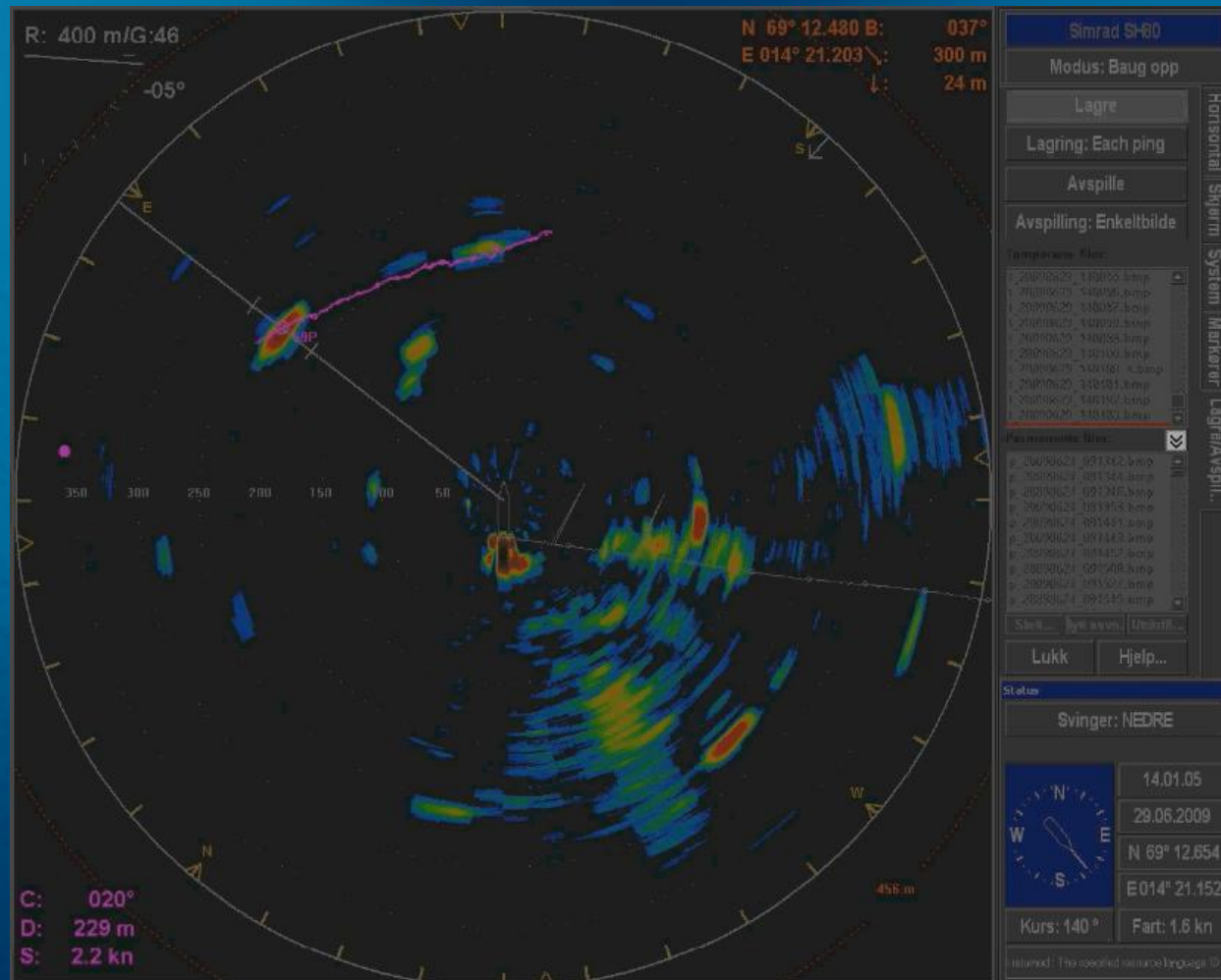
Examples of sonar data

SX90 data of large herring school 2012



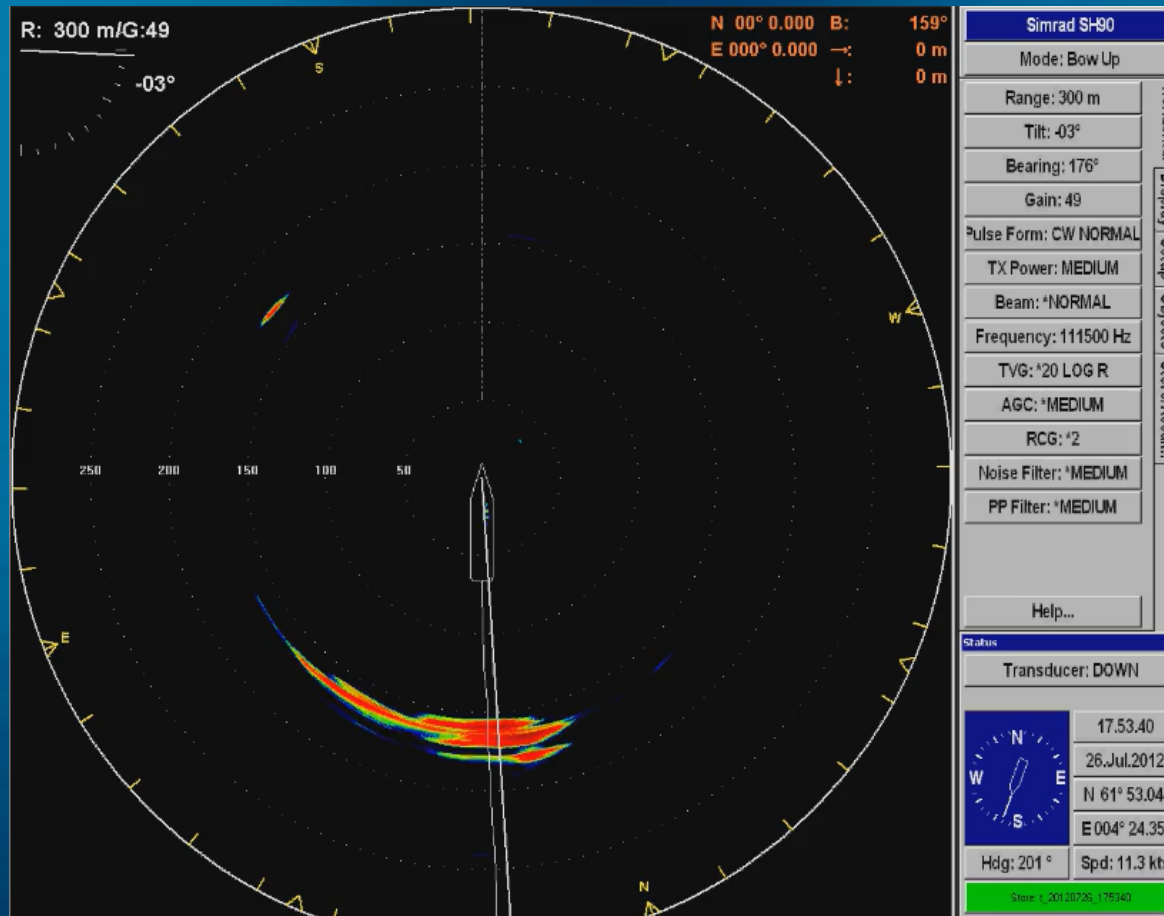
Examples of sonar data

SX90 data of small herring school, vessel drifting 2009



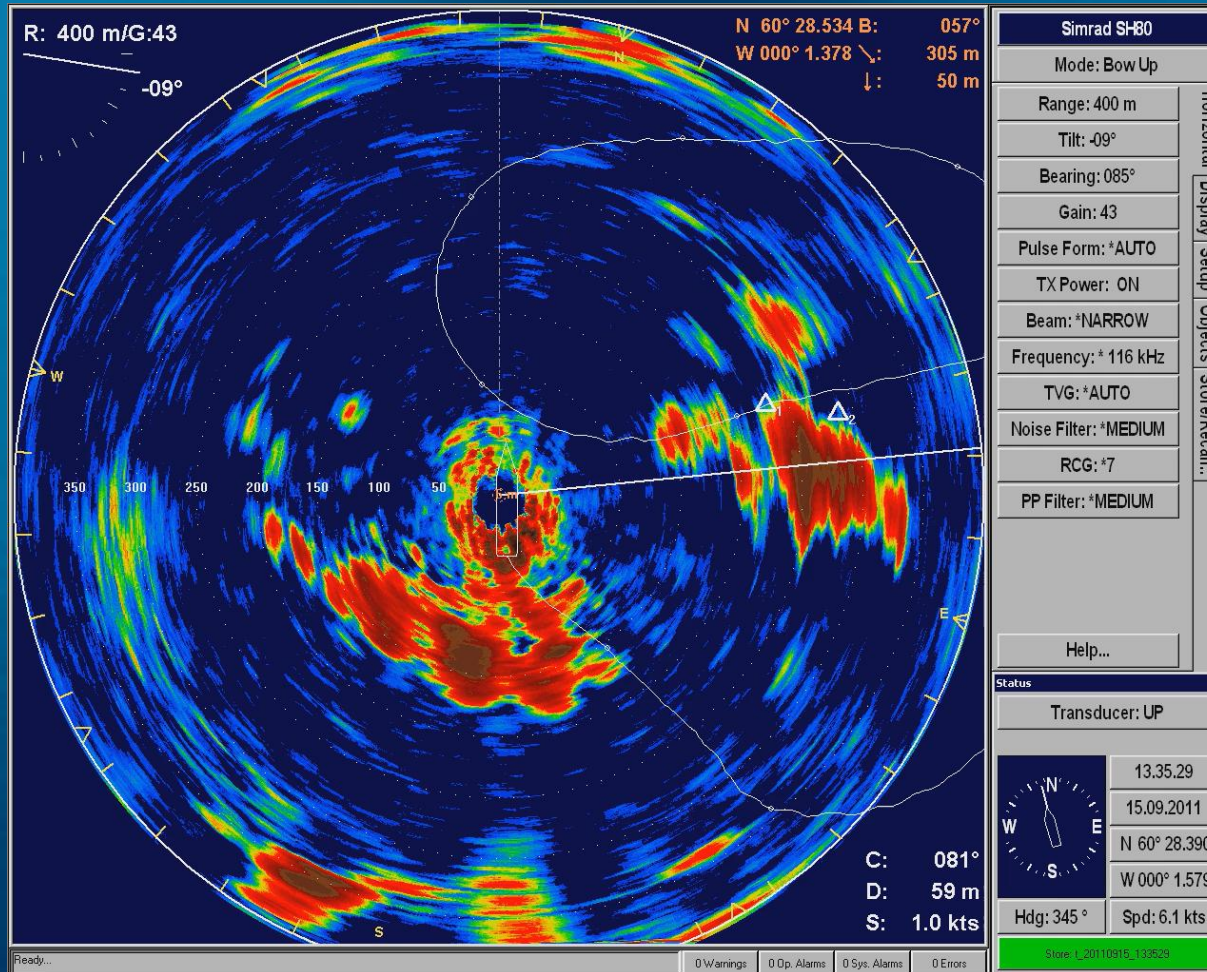
Examples of sonar data

SH90 data of small mackerel school, surveying 2012



Examples of sonar data

SH80 data of large mackerel school, fishing 2011



Examples of sonar data

SX90 data of large herring school inside a fjord, 2009

